CIS 5200 Fall 2025, Project Report

Project Title: ECG Image Digitization

Team Name: Team Unsupervised

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Abstract

This project addresses the critical problem of Electrocardiogram (ECG) image digitization, aiming to automatically extract the underlying 12-lead time-series signals from scanned or photographed paper printouts. To overcome challenges posed by real-world image artifacts, grid patterns, and noise , we decomposed the task into a robust three-stage machine learning pipeline : Stage 0: Global Geometric Normalization, Stage 1: Local Non-linear Rectification, and Stage 2: Waveform Extraction. We systematically evaluate three methods, including a feature-driven Context-Aware Random Forest baseline, an end-to-end Residual 1D CNN, and a novel Baseline-Aware Binary Segmentation Network, with an extension reproducing and comparing the Large-Scale Coordinate-Augmented ResUNet. Comparative analysis reveals that deep learning segmentation architectures, which treat the problem as pixel-level dense prediction, significantly outperform regression-based models in terms of signal fidelity (SNR), with the extension ResUNet achieving an optimal SNR of 16.42 dB. Our results confirm that the structured, multi-stage approach is essential for stable optimization, and that pixel-wise segmentation is superior to sequence-to-sequence regression for this task. We identify that the primary limitation of our segmentation model is the reconstruction of high-frequency structures (QRS complexes) due to the limited scale and class imbalance of the real training dataset. Future work is therefore proposed to use massive synthetic data augmentation and refined loss functions to improve high-fidelity signal extraction. The Code for this project can be found on Github

1 Motivation

In the field of medical, vast archives of past patient data are still stored in non-digital formats, such as paper printouts and scanned images. This large amount of information is inaccessible to modern computational analysis, data mining, and the development of large-scale machine learning models. Electrocardiograms (ECGs), one of the most fundamental and common diagnostic tools for cardiac health, is a perfect example of this problem. Globally, there are billions of old ECG records that are only available as paper charts or digitized images. The critical time, series signal data, the real diagnostic information, cannot be algorithmically examined or utilized for research purposes, even though they have been digitized as pictures.

This project aims to solve this critical data-unlocking problem. Our goal is to create a machine learning pipeline that can automatically "digitize" scanned ECG images and precisely extract the underlying 12-lead time-series signals. This task is challenging for traditional computer vision methods, which struggle with the considerable variety in scan quality, background grid patterns, image noise, and handwritten artifacts. Machine learning, particularly deep learning, is uniquely suited to this challenge as it can learn to robustly identify complicated spatial patterns, distinguishing the signal traces from background noise and distortions in a way that rule-based systems cannot. Successfully digitizing this data would open up enormous, priceless databases for cardiovascular research and the creation of next-generation diagnostic artificial intelligence.

2 Related Work

The problem of ECG image digitization has been studied for more than a decade, with earlier approaches relying primarily on classical image-processing techniques. Traditional methods typically used global or adaptive thresholding, morphological filtering, and line-detection algorithms such as the Hough Transform to isolate waveform traces from background grids [2]. While effective on clean, uniformly scanned images, these rule-based approaches lack robustness to real-world artifacts, including inconsistent paper quality, nonuniform lighting, skewed scanning, handwritten annotations, and the dense red gridlines commonly found in clinical ECG printouts. Their reliance on handcrafted features severely limits generalization beyond carefully curated datasets.

With the rise of deep learning, more recent work has shifted toward convolutional neural networks and semantic-segmentation architectures capable of learning spatial representations directly from data. U-Net and its variants have been widely explored for extracting ECG traces from noisy backgrounds [5]. These models significantly outperform classical methods, but they still face difficulties when the signal lines are extremely thin, discontinuous, or overlapped with grid artifacts. Hybrid approaches have therefore emerged as a promising direction. For example, the SignalSavants team, the winners of the 2024 George B. Moody PhysioNet Challenge that combined the Hough Transform for geometric normalization with a deep neural network for fine-grained waveform extraction [4]. Their success highlights that robust digitization requires both reliable preprocessing and powerful learned representations. Building upon insights from prior work and recent PhysioNet challenges [1], our project aims to systematically evaluate multiple ML and DL architectures to understand their strengths, limitations, and applicability to real-world ECG image-to-signal conversion.

3 Dataset

We will use the **PhysioNet ECG Image Digitization** dataset from the official Kaggle competition (https://www.kaggle.com/competitions/physionet-ecg-image-digitization). The dataset is designed for the task of reconstructing digital electrocardiogram (ECG) timeseries signals from scanned or photographed images of paper ECG printouts. It provides a realistic benchmark for developing robust computer vision and signal processing models that can digitize legacy clinical ECGs.

Data composition. Each sample consists of:

- A high-resolution image of a 12-lead ECG printout (`.png` or `.jpg`);
- The corresponding raw waveform signals stored in the WaveForm DataBase (WFDB) format, including a header file (`.hea`) describing the recording metadata and a binary signal file (`.dat`) containing the multi-channel voltage traces;
- Optional metadata describing sampling frequency, amplitude scaling, and patient or device identifiers.

Dimensions. The dataset contains approximately $n \approx 1000$ image-signal pairs across training, validation, and test splits. Each ECG record includes $p = 12$ leads (channels), typically sampled at 500 Hz with durations of 10-15 seconds, resulting in roughly 5,000-7,500 time points per channel. Images vary in resolution (roughly 1000–3000 pixels width) and exhibit real-world variations such as gridline artifacts, folds, scanning noise, and illumination differences.

Feature structure. The input features for our models will be pixel-level image intensities (after preprocessing such as contrast normalization and background removal), while target outputs are continuous voltage values representing the digitized ECG waveform for each lead. Potential auxiliary features include extracted morphological descriptors (e.g., QRS width, RR interval) and spectral representations used for downstream evaluation.

4 Problem Formulation

We formulate ECG image digitization as a *supervised structured regression* problem, where the objective is to map a printed ECG image X into a collection of continuous lead-specific time-series signals Y. Each sample consists of an input image $X \in \mathbb{R}^{H \times W \times 3}$ and corresponding ground-truth waveforms $\{y^{(l)}\}_{l=1}^{L}$, where $y^{(l)} \in \mathbb{R}^{T_l}$ denotes the numerical ECG signal for lead l.

To make the learning problem tractable, we decompose it into three sub-tasks aligned with the visual structure of ECG images.

4.1 Preprocessing

Image Rectification (Stage 0). We treat orientation correction and layout normalization as a keypoint regression problem, implemented using a pretrained convolutional neural network [4] to predict geometric or layout keypoints (K). From these predicted keypoints, we estimate a homography that maps each distorted ECG image into a standardized coordinate frame. This pretrained-model formulation is effective because even small rotation or perspective deviations propagate into waveform extraction errors; enforcing geometric consistency therefore provides a stable and reliable foundation for all downstream learning tasks.

Grid Detection (Stage 1). We frame gridline localization as a dense landmark regression task, implemented using a pretrained neural network[4] to detect grid intersections and spacing patterns. Accurate grid detection establishes a pixel-to-time and pixel-to-voltage coordinate system, enabling the conversion of pixel-level waveform predictions into continuous physiological values. This step functions as an explicit form of feature engineering: rather than relying solely on raw pixels, the model leverages a structured coordinate representation that simplifies and stabilizes the subsequent regression tasks.

4.2 Waveform Extraction (Stage 2)

Given the rectified image, waveform extraction becomes a *pixel-wise regression* problem in which the model predicts the vertical location of the ECG trace for each horizontal time coordinate. The predicted pixel series is then transformed into a numerical waveform using grid spacing and baseline calibration. We use an L_2 reconstruction loss

$$\mathcal{L} = \sum_{l=1}^{L} \|y^{(l)} - \hat{y}^{(l)}\|_2^2,$$

which is appropriate because the evaluation metric for the digitization task is also based on point-wise signal accuracy.

This three-stage formulation reflects the highly structured nature of the mapping from image pixels to ECG signals: it involves geometric alignment, grid-based coordinate decoding, and waveform reconstruction. A single end-to-end black-box regression performs poorly in preliminary experiments and in prior work, whereas decomposing the task into interpretable sub-modules yields more stable optimization and more explainable intermediate representations.

5 Methods

5.1 Preprocessing Pipeline (Stages 0 and 1)

The entire pipeline, *adopted from the work of Krones et al.* [4], ensures the ECG images meet the strict linear mapping requirement for signal extraction, mitigating both global and local geometric distortions.

5.1.1 Stage 0: Global Geometric Normalization

The initial stage addresses global rotation and perspective distortion. A ResNet-18 encoder [7] extracts multi-scale features, which are then used to predict lead position keypoints via a U-Net decoder. The final

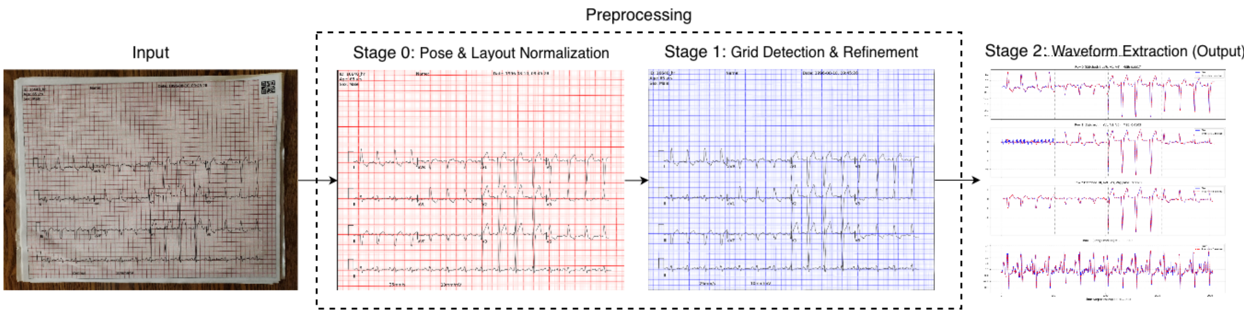

Figure 1: Flowchart of the preprocessing pipeline.

keypoints are refined using Test-Time Augmentation (TTA) and extracted from the heatmap via connected-component analysis [6]. A RANSAC-based homography estimation [3] subsequently aligns the entire image into a canonical coordinate system (1152 $\times$ 1440 resolution), providing a stable foundation for downstream tasks.

5.1.2 Stage 1: Local Non-linear Rectification

The primary objective of this stage, following the methodology of Krones et al. [4], is to eliminate local, non-linear geometric distortions (e.g., paper wrinkles) that persist after Stage 0. We employ a U-Net-like `Stage1Net` network within a Multi-task Learning (MTL) framework for dense prediction. This approach simultaneously predicts three geometric properties: a probability map for Gridpoint Prediction, and pixel-wise segmentation for Horizontal and Vertical Line Segmentation. Post-processing uses Connected Component Analysis (CCA) to extract a precise coordinate matrix (S) representing the distorted paper mesh. A non-linear mesh warping operation is then performed to align these detected grid points with an ideal Euclidean reference grid (T), yielding the final geometrically standardized "rectified" image required for Stage 2.

5.2 Method 1: Baseline - Context-Aware Random Forest

The Context-Aware Random Forest baseline is designed as a feature-driven regression method that estimates ECG voltage values directly from the rasterized grayscale ECG image. The core idea is to transform each image column into a compact, 15-dimensional descriptor that captures the vertical distribution and local morphology of the ECG trace. These descriptors include statistical heuristics such as the weighted center of mass and dispersion, supplemented by features from adjacent columns to incorporate short-range spatial continuity. Once extracted, the resulting feature sequence is passed through a Random Forest Regressor, which models the nonlinear mapping from image-derived statistics to the underlying voltage waveform. A Savitzky-Golay filter is applied to the predicted waveform to enforce temporal smoothness, followed by a zero-centering correction to remove global drift. This structured, non-deep learning approach provides an interpretable baseline against which the proposed neural architectures can be compared.

5.3 Method 2: Residual 1D CNN

An end-to-end framework combining a 2D convolutional encoder with a 1D convolutional decoder was developed to reconstruct multi-lead ECG waveforms from rasterized ECG images. This approach is motivated by the observation that ECG images exhibit a natural temporal structure along the horizontal axis while vertically encoding voltage variations. A 2D CNN is therefore well suited for capturing the local shapes of ECG components, such as QRS complexes and P/T waves, while a 1D CNN provides a principled way to model temporal dependencies in the reconstructed signal. We employ a pretrained ResNet-34 from the timm [7] library as the encoder, leveraging its strong feature extraction capability and stable training dynamics. The encoder outputs a high-dimensional spatial feature map, which is passed through a 1×1 convolution for channel reduction and an adaptive average pooling layer to collapse the vertical dimension. This produces a temporally ordered feature sequence aligned with the target waveform domain. To reconstruct the ECG signal, we design a lightweight 1D convolutional decoder composed of several Conv1d and ReLU layers, concluding with a 1×1 Conv1d to produce the final multi-lead output. We choose 1D convolutions due to their efficiency, inductive bias for local temporal patterns, and robustness when modeling moderate-range temporal dependencies. The entire architecture is implemented in PyTorch, which provides modular components and automatic differentiation, enabling fully end-to-end optimization of the mapping from images to voltage sequences.

5.4 Method 3: Baseline-Aware Binary Segmentation Network

To reconstruct the ECG traces with high spatial precision, we design a baseline-aware U-Net architecture that predicts four binary segmentation maps corresponding to the ECG channel groups. The network uses a pretrained ResNet18 encoder with a four-stage decoder, followed by a classification head that outputs a $4 \times H \times W$ probability map. To provide explicit spatial context, we concatenate three types of embeddings to the decoder features: (1) a normalized y-coordinate map, (2) distance-to-baseline embeddings that encode each channel's 0 mV reference line, and (3) soft channel-indicator maps computed from Gaussian proximity to each baseline. These embeddings allow the model to disambiguate vertically stacked ECG leads and suppress false positives far from the appropriate baseline. The loss function combines Dice, Focal, and BCE terms with an additional spatial penalty that discourages predictions inconsistent with baseline geometry. This baseline-conditioned segmentation provides a robust, deterministic representation that enables high-fidelity signal extraction in subsequent processing.

5.4.1 Prior Method Comparison: Large-Scale Coordinate-Augmented ResUNet

As a point of reference for evaluating our baseline-aware formulation, we include a waveform-extraction method developed by others[4]. Due to its accurate, high-fidelity signal reconstruction, it is used as a comparative method to analyze the potential improvements of out baseline-aware binary segmentation network method. The approach trains a ResNet-nnUNet segmentation network to predict pixel-level masks that indicate the locations of ECG waveforms. To supervise the model, a large synthetic dataset is constructed from PTB-XL time-series signals using the ECG-Image-Kit framework. Each ECG is rendered into an image with synthetic grids and appearance variations (color shifts, wrinkles, shadows, annotations) and paired with a precisely generated binary mask marking the waveform pixels.

6 Experiments and Results

6.1 Experiment with different methods

6.1.1 Baseline - Context-Aware Random Forest

The Random Forest model is implemented using scikit-learn, initialized with 50 estimators and a maximum depth of 10, which we found to prevent overfitting while maintaining sufficient representational capacity. The model is trained using the default mean-squared-error objective. All predictions undergo post-processing using a Savitzky-Golay filter (window size 51, polynomial order 3) to smooth high-frequency artifacts, after which waveforms are normalized by subtracting the global mean. The result is shown in Figure 2.

6.1.2 Residual 1D CNN

As for residual 1D CNN, all experiments are implemented in PyTorch and follow a consistent data split, allocating 80% of samples to the training set and the remaining 20% to the validation set. Input ECG images are uniformly resized to 512×2048 pixels and normalized using the same preprocessing pipeline for all methods to ensure comparability. Model training is performed using the AdamW optimizer with a learning rate of 1×10^{-4} and standard weight decay. A mini-batch size of 12 is applied, and automatic mixed precision (AMP) is enabled to reduce GPU memory consumption and accelerate training. Each model is trained for three epochs, with validation performed at the end of every epoch. The result is shown in Figure 3.

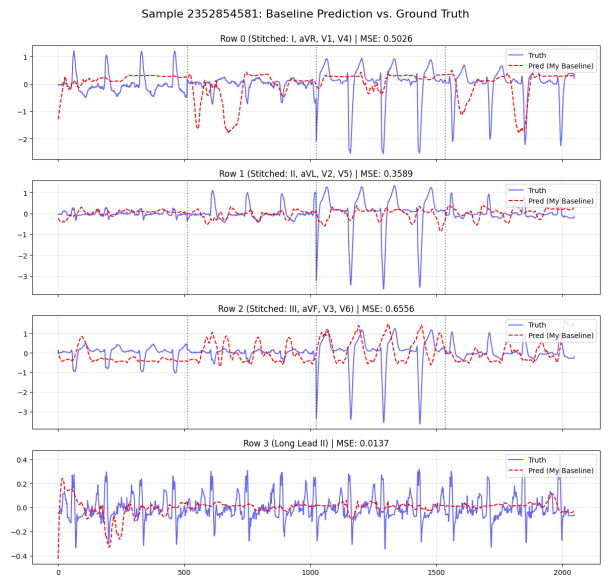

Figure 2: Baseline method Context-Aware Random Forest sample prediction

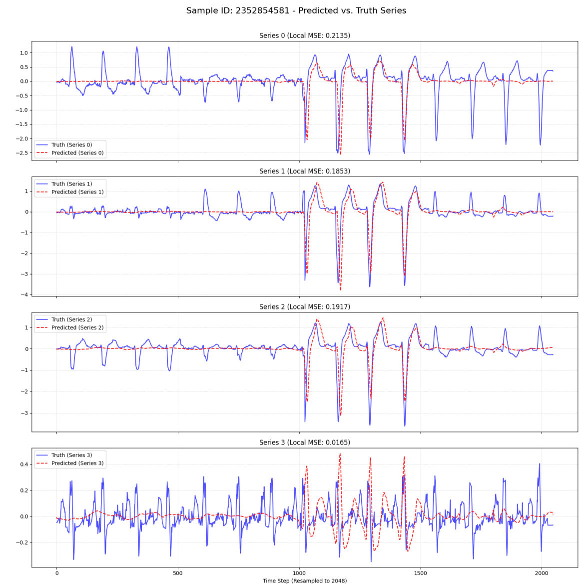

Figure 3: Method 2 Residual 1D CNN sample prediction

6.1.3 Baseline-Aware Binary Segmentation

We adopt a reproducible setup using a fixed random seed (42) for all train/validation splits and data shuffling. Our model is a ResNet18-based U-Net initialized with ImageNet-pretrained weights. Training uses AdamW with a learning rate of 1×10^{-4}, weight decay 5×10^{-4}, and a `ReduceLROnPlateau` scheduler (factor 0.5, patience 3, minimum LR 1×10^{-6}). We train for up to 100 epochs with early stopping (patience 15), saving checkpoints every five epochs and on the best validation score. The loss is a weighted combination of Dice, Focal, and BCE losses (2.0/0.5/0.1), plus auxiliary SNR (0.1) and MSE (0.05) terms. Mixed precision is enabled, with batch size 1 (GPU) or 2 (CPU). Data augmentation includes horizontal flips (50%), brightness jitter (0.85–1.15), and Gaussian noise ($\sigma = 3$). The dataset is split 80/20 for training and validation. The result of baseline-aware binary segmentation is shown in Figure 4 and the sample result of the large-scale coordinate-augmented ResUNet is shown in Figure 5.

6.2 Evaluation Framework

For reproducibility, we adopt a fixed train-test allocation in which 80% of the valid ECG images are reserved as the training set, and the remaining 20% are allocated for evaluation. To ensure consistent comparisons across all samples, evaluation is performed against a Ground Truth (GT) waveform reconstructed from the raw CSV files using a stitching procedure that concatenates the corresponding 2.5-second lead segments into a unified 10-second signal. Both GT and predicted waveforms are uniformly interpolated to 2048 sample points to eliminate differences in sampling density. This alignment ensures that performance metrics reflect true temporal correspondence between predictions and the physiological signals.

6.3 Evaluation Metrics

The performance of the ECG image digitization pipeline is evaluated using a dual-metric approach, reflecting both the training optimization needs and the ultimate goal of high signal fidelity required by the competition.

1. Signal-to-Noise Ratio (SNR) The Signal-to-Noise Ratio (SNR) measures the quality of the digitized signal ($\mathbf{V}_{\text{pred}}$) relative to the error (noise) introduced by the digitization process. A higher SNR value

Figure 4: Method 3 Binary Segmentation sample prediction

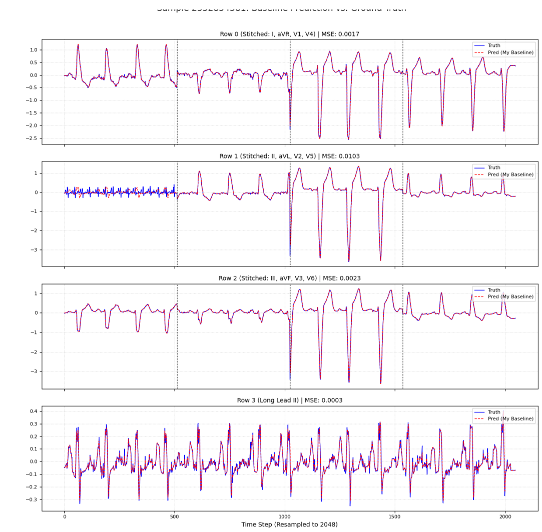

Figure 5: Large-Scale Coordinate-Augmented ResUNet sample predictionn

indicates a lower level of noise, confirming higher fidelity and accuracy.

The SNR is calculated in decibels (dB) as:

$$\text{SNR} = 10 \log_{10} \left(\frac{\sum_{i=1}^{N} \mathbf{V}_{\text{true},i}^2}{\sum_{i=1}^{N} (\mathbf{V}_{\text{true},i} - \mathbf{V}_{\text{opt},i})^2} \right)$$

where:

- N is the number of sampled points.
- $\mathbf{V}_{\text{true}}$ is the ground truth signal.
- $\mathbf{V}_{\text{opt}}$ is the predicted signal ($\mathbf{V}_{\text{pred}}$) optimally shifted in time via cross-correlation to achieve the best possible alignment with $\mathbf{V}_{\text{true}}$.

2. Mean Squared Error (MSE) The Mean Squared Error (MSE) measures the average squared difference between the predicted and true voltage values. It is a smooth, differentiable, and convex function, making it ideal for the backpropagation process in deep learning model training. MSE is defined as:

$$\text{MSE} = \frac{1}{N} \sum_{i=1}^{N} (\mathbf{V}_{\text{true},i} - \mathbf{V}_{\text{pred},i})^2$$

where N is the number of sampled points.

6.4 Results

Table 1: Performance comparison of four methods

Method	**MSE** (mV^2)	**SNR** (dB)
Baseline: Context-Aware Random Forest	0.11	-10.92
Residual 1D CNN	0.034	0.69
Baseline-Aware Binary Segmentation	0.276	5.57
Reference Prior: Large-Scale Coordinate-Augmented ResUNet	0.0014	16.42

7 Conclusion and Discussion

We intend to summarize the comparative performance of the different models, analyzing the trade-offs between computational complexity and signal fidelity. Furthermore, we will discuss specific failure cases, such as noisy or distorted images, and propose potential improvements for future work.

A direct comparison reveals that methods relying on pixel-level dense prediction (Binary Segmentation and ResUNet) significantly outperform the regression-based approaches (Random Forest and Residual 1D CNN) in terms of signal fidelity (SNR). This strongly suggests that treating ECG digitization as a pixel-wise segmentation problem is superior to sequence-to-sequence regression for this specific structured input. The **Reference Prior method** achieves the optimal result, solidifying the choice of a U-Net-like architecture for this task.

The Random Forest baseline offers a simple and interpretable alternative to deep learning, but its performance is fundamentally constrained by the handcrafted 15-dimensional features extracted from each image column. These descriptors capture only coarse statistical properties and cannot represent the fine, thin, and high-curvature morphology of ECG traces, especially around QRS complexes. Moreover, treating columns independently prevents the model from learning temporal structure or correcting for local ambiguities caused by gridlines and noise. As a result, the model produces oversmoothed waveforms and achieves the lowest SNR and highest MSE among all methods. Nevertheless, it provides a useful lower bound that illustrates the necessity of learning spatial and temporal patterns directly from the image domain using deep neural networks.

The Residual 1D CNN successfully demonstrates the benefits of a deep learning approach over the traditional feature-driven Random Forest baseline, achieving a lower MSE and a much higher SNR. This confirms that deep learning models are better suited for capturing the complexity of ECG waveforms than relying on simple, handcrafted statistical features. However, the Residual 1D CNN's end-to-end framework, which collapses the vertical dimension through average pooling to produce a temporal feature sequence , fundamentally simplifies the complex, vertically-stacked, multi-lead structure of the ECG image. While the 1D CNN captures temporal dependencies, it may struggle with the high-precision, pixel-level regression needed to accurately map the distinct vertical locations of the 12 leads, which is a key advantage of the U-Net's dense, pixel-wise prediction capabilities. Therefore, while it outperforms the feature-constrained baseline, it reveals that a segmentation/pixel-prediction architecture is more effective for the highly structured, multi-channel ECG digitization task.

A key distinction between our baseline aware binary segmentation method and the reference large scale ResUNet approach lies in the training data regime. Our baseline-aware U-Net is trained on fewer than 1,000 real ECG images, whereas the prior method relies on tens of thousands of synthetically generated ECG printouts with perfectly rendered pixel-level masks. This disparity has a direct impact on model behavior: our network learns the smoothly varying portions of the waveform that dominate the training distribution and are tightly coupled to the baseline geometry, but it struggles to faithfully reconstruct sharp deflections such as QRS complexes. These spikes constitute an extreme minority class in pixel space, and without large-scale synthetic examples covering their variation, the segmentation loss is dominated by background and near-baseline pixels. Consequently, the model becomes highly effective at baseline-adjacent tracking but exhibits reduced sensitivity to high-frequency, low-support structures. This observation highlights the importance of dataset scale and class balance for pixel-level ECG digitization and motivates future work incorporating controlled synthetic augmentation to improve spike fidelity.

The overall project provided valuable lessons in the workflow of machine learning research, particularly for structured computer vision problems. We learned to match the method to the problem structure. The most effective approach included moving from an interpretable baseline (Random Forest) to test the deep learning hypothesis (1D CNN) , and finally confirming that a Segmentation architecture (U-Net) was the optimal choice for the Structured Regression nature of the task. The initial attempts at pure end-to-end regression were ineffective, while the three-stage formulation, Pose Normalization (Stage 0), Grid Detection (Stage 1), and Waveform Extraction (Stage 2), was critical for success. This process transforms the complex image-to-signal mapping into a series of stable, interpretable sub-modules. We also realized that beyond choosing the correct methodology, computational resources are a limiting factor, especially when aiming to match state-of-the-art results. The Reference Prior ResUNet is a large-scale model trained on massive synthetic data, which requires substantial GPU resources and training time. Our resource constraints limited our ability to: (1) use a larger U-Net backbone, (2) fully leverage large batch sizes, and (3) train for a very large number of epochs. Simply choosing the right method is insufficient; scaling both the dataset and the model capacity is equally important.

Based on our findings and the comparative performance analysis, we propose two primary opportunities for future research to improve the ECG image digitization pipeline. The most critical extension involves Massive Synthetic Data Augmentation, aiming to replicate the success of the high-performing reference method by generating a large-scale synthetic dataset containing tens of thousands of samples. This approach will balance the training distribution and significantly improve the fidelity of reconstructing minority class structures, such as the sharp QRS complexes. Secondly, we recommend Loss Function Refinement for High-Frequency Structures. This involves developing a custom loss function that assigns a higher weight or penalty to errors associated with high-frequency, sharp features (e.g., based on a signal's curvature or second derivative), which will directly address the model's current weakness in accurately reconstructing spikes.

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